Problem B

Design and Dynamic Analysis of a Roller Coaster

Team # 389

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Abstract

This paper analyses the track and motion of a roller coaster.Our main task can be divided into three parts:

(1)Design a roller coaster that meets both security and stimulation.

We make the following analysis: While riding a roller coaster, however, being overweight or weightless and changing direction can cause changes in acceleration, resulting in a sense of excitement. However, excessive acceleration is bound to cause damage to the health of passengers. Normal adults can withstand 4~5g vertical acceleration with a duration of no more than 0.1s.According to the theoretical calculation, the maximum vertical acceleration in this design scheme is 4.1g, and the duration is 0.01s, which occurs at the first downhill. The rest of the vertical acceleration is less than 4g. The time of overweight and weightlessness accounts for 87% of the total time. Therefore, this design scheme can continuously bring passengers a sense of tension and stimulation under the premise of ensuring safety.

(2)Analyze the dynamics of roller coaster and calculate the total duration.

The roller coaster track is divided into 9 segments. In each segment, we use Kinetic energy theorem to analyse its motion, and then obtain velocities between every two adjacent segments. These velocities correspond well with results of simulation. With the help of simulation softwares, we figure out the total duration of the ride is 1 minute and 2.67seconds.

(3)Accurately describe the motion trajectory of roller coaster in three dimensions.

According to (1) and (2), we can precisely describe the path that the roller coaster track would follow.

Keywords: Roller coaster, Simulation, Kinetic energy theorem, Dynamical analysis

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1. Problem Analysis

1.1 Introduction

Roller Coaster is an exciting activity, which brings riders a lot of fun, excitement and thrill. Aboard on a roller coaster, the sense of weightless and overweight when the roller coaster up and down is why we feel excited of thrilled. However, there are some accidents causing injuries or deaths, a few of which may result from some unreasonable design. What's more, the up-and-down roller coaster could possibly make us feel sick if the sense of acceleration is too excessive. In this paper, we try to analyse how a roller coaster works and try to design a roller coaster safe enough as well as exciting. We design different parts of its path, including riser, vertical ring, arc segments and so on. The dynamic analysis of each part will be posted in this paper.

1.2 Problem Analysis

The roller coaster is required to start from a horizontal section of 20 meters and rise to the highest point of a mountain under the action of external forces. The height of the highest point is 30 meters, and its gravitational potential energy is the only energy source of the roller coaster motion. The roller coaster track should be closed and not

lower than the initial height.

Our main **task** can be divided into three parts:

(1) Design a roller coaster that meets both security and stimulation.

(2) Analyze the dynamics of roller coaster and calculate the total duration.

(3) Accurately describe the motion trajectory of roller coaster in three dimensions.

Firstly, we start with roller coaster track design. Through consulting materials, we know that the track of roller coaster is usually combination of a series of mechanical models, which make the car body climb, fall, reverse and other actions along the track. Track systems should be considered for safety and stimulation. We make the following analysis: roller coaster stimulation mainly depends on the acceleration, through the appropriate overweight, weightlessness, impact, vibration and other sensory stimulation to the passengers. However, excessive acceleration will cause harm to human body, if affect vestibular organ, make human body produces the undesirable physiology reaction such as dizziness, vomiting. Accelerated passage can also cause stress on the heart, blood pressure, etc.

Therefore, we must come up with a feasible plan to keep the roller coaster's acceleration within a certain range, which is both

exciting and safe. We use the combination of clothoid loop, circular arc track and spiral track to design a roller coaster track system that meets the requirements.

According to the track model, we will analyze the force and motion of each part of the car, and complete the dynamic analysis and the solution of the total motion time with the help of the simulation software.

Combined with the above dynamic analysis, we can accurately describe the motion process of roller coaster in three dimensions.

2.Model

2.1 Model assumption

This model is established on the basis of actual roller coaster track. In order to facilitate theoretical modeling and calculation, some practical conditions need to be limited and some minor parameters need to be reasonably simplified. Based on this model, the following assumptions are made.

[Assumption1]

When calculating wind resistance, only the windward of roller coaster is considered, not the transverse wind resistance.

[Assumption2]

The roller coaster started at the top of the mountain with an initial

speed of zero.

[Assumption3]

The wheel of the car is only subject to rolling friction. The maximum rolling friction moment is used to calculate the power consumption in the process of motion. The rolling friction coefficient of the track is equal everywhere.

[Assumption4]

Roller coasters pass through certain small arcs that are approximately free of drag (wind resistance and rolling friction).

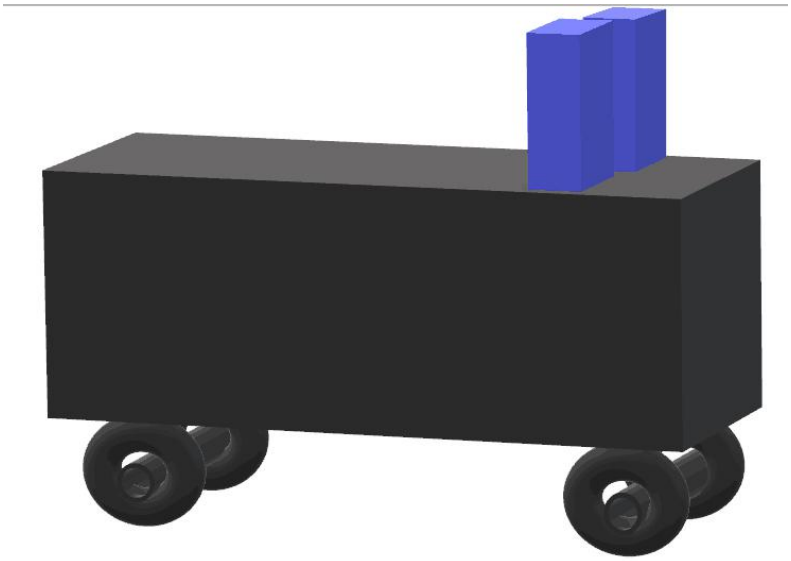


Figure 1 carriage

[Assumption5]

The carriage wheels are upright.

2.2 Parameter setting

The empirical value of some parameters can be obtained by referring to the data, which can be directly substituted into the theoretical

derivation as a known quantity.

See the table below for specific values.

Table 1 constant

Symbol	Physical meaning
$r_i \left(i = 1, 2, \ldots, 5\right)$	Radius of arcs
$\mathrm{v}_i(\mathrm{i} = 1, 2, \ldots, 10)$	The initial or final velocity of every section
$\mathrm{v}_j(\mathrm{j} = \mathrm{I}, \mathrm{II}, \ldots, \mathrm{VI})$	The average velocity of every section
$W_i(\mathrm{i} = 1, 2, \ldots, 7)$	The resistance work of every section
$l_i(\mathrm{i} = 1, 2)$	The length of every straight section
$h_i(\mathrm{i} = 1, 2)$	The height of every parts of the riser
F_w	The resistance work of air-resistance
f	The rolling friction between the wheels of the roller coaster and the track
ρ	The indensity of air
A_D	The windward area
C_D	Air resistance coefficient
K	Rolling friction coefficient
s	arc length
θ	Variable angle
α	The specific angle
m	The total mass of passengers and carriages

The unknowns involved in the model are shown in the table 2.

Table 2 variable

symbol	value
r_1	3 meters
r_2	5 meters
r_3	5 meters
r_4	50 meters
r_5	30 meters
l_1	15 meters
l_2	30 meters
ρ	1.3 kilograms per cubic meter
A_D	1.5 square meters
C_D	0.3
K	0.05
g	9.8 newton per meter
α	$\frac{\pi}{6}$
m	2.2 ton

2.3 Model Establishment

Using Nolimts II: Roller Coaster Simulation software for the design of a Roller Coaster model kinematics Simulation, first of all in strict accordance with the design specifications of a Roller Coaster

track modeling, using block design train of thought, each segment will orbit parameters such as dimension, Angle and height of the input, the orbital diagram.Here are the views of five directions.

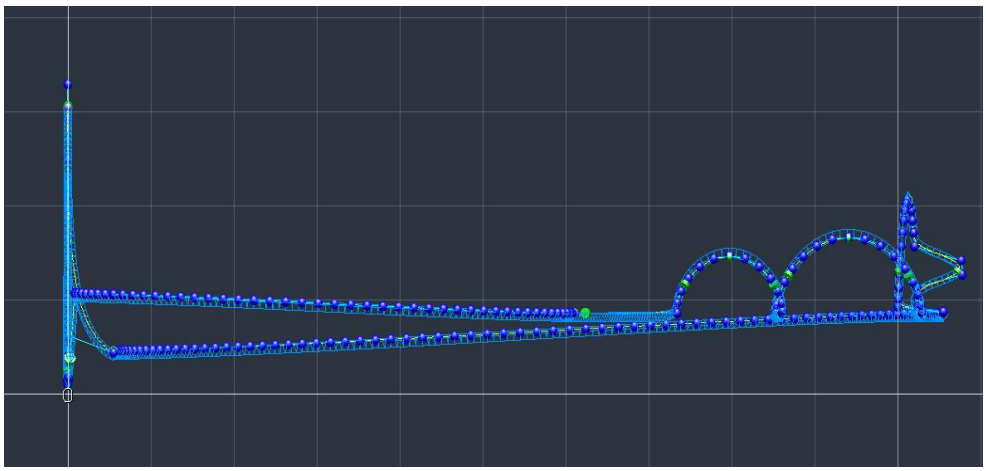


Figure 2 front view

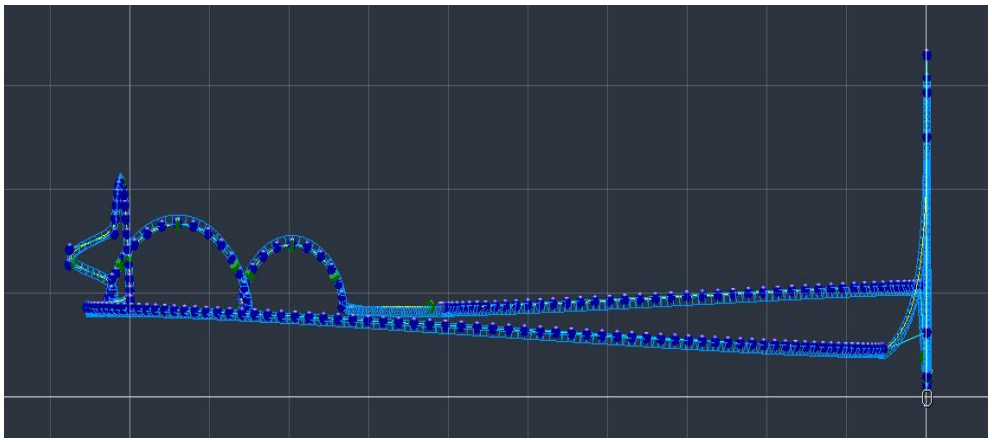


Figure 3 rear view

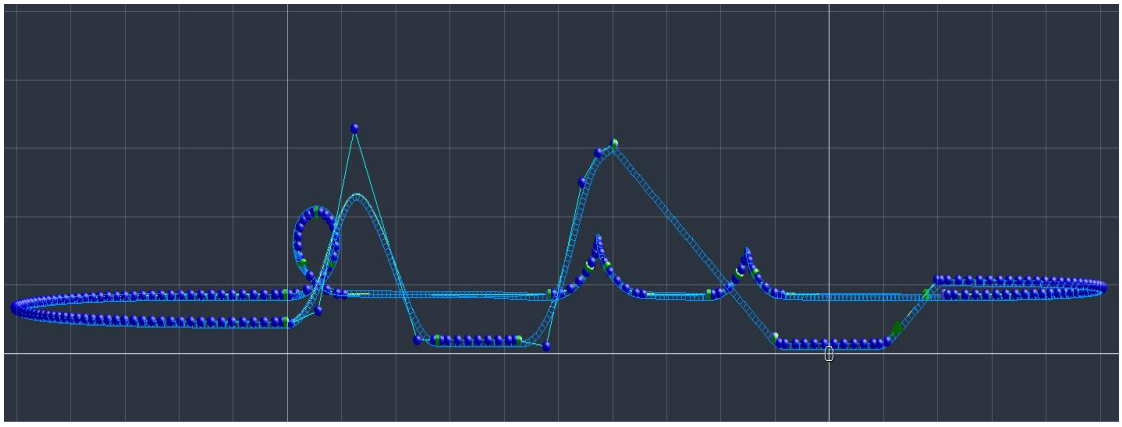


Figure 4 left view

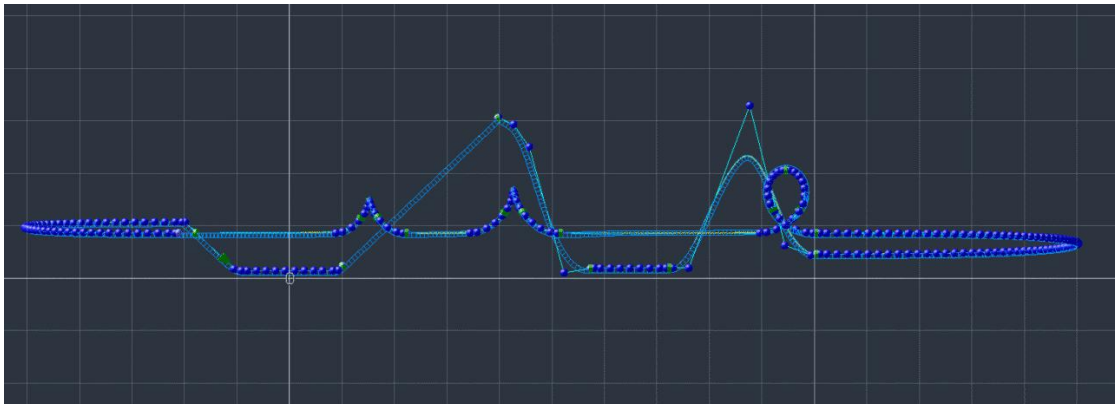


Figure 5 right view

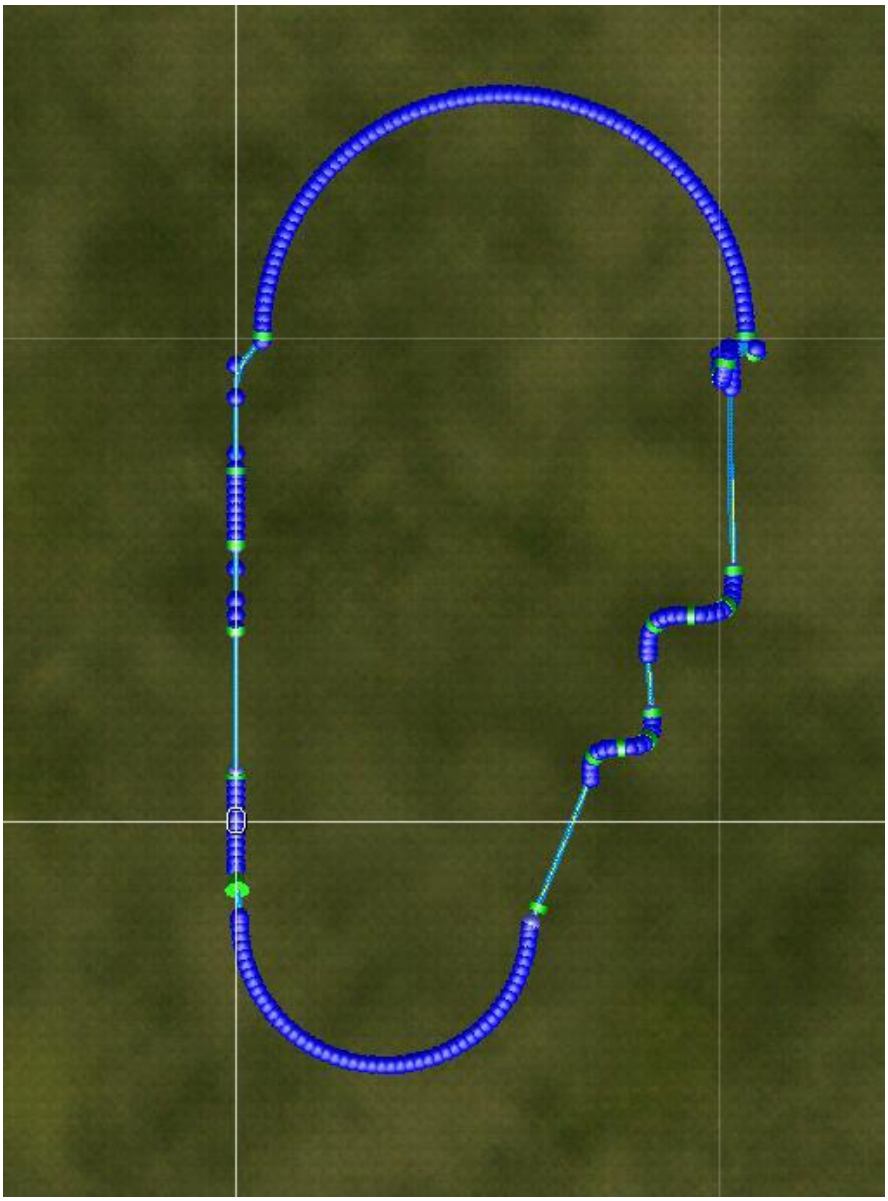


Figure 6 top view

2.4 Formula Derivation

In accordance with kinetic energy theorem, there have

$$W_{gravity} - W_{resistance} = \frac{1}{2}mv_t^2 - \frac{1}{2}mv_o^2 \tag{1}$$

$W_{gravity}$ is gravitational potential energy, $W_{resistance}$ is consumed energy against resistance, m is the total mass of passengers and carriages, v_t is final velocity, v_o is initial velocity.

Based on the theory of aerodynamics, The wind resistance formula can be expressed as

$$F_w = \frac{1}{2}\rho A_D C_D v_j^2 \tag{2}$$

F_w is wind resistance, ρ is the indensity of air($1.3kg/m^3$), A_D is the windward area($A_D = 1m \bullet 1.5m = 1.5m^2$), C_D is Air resistance coefficient(0.3), v_j is The average velocity of every section.

Especially

$$v_j = \frac{v_i + v_{i+1}}{2} \tag{3}$$

According to [Assumption3], The rolling friction

$$f = KN \tag{4}$$

total friction

$$F_{resistance} = F_w + f \tag{5}$$

Then,consumed energy against resistance

$$W_{resistance} = \int F_{resistance} ds = \int (F_W + f) ds \tag{6}$$

simultaneous equations (1)~(6),acquire the **model 1**

$$\begin{cases} W_{gravity} - W_{resistance} = \frac{1}{2}mv_t^2 - \frac{1}{2}mv_o^2 \\ W_{resistance} = \int F_{resistance} ds = \int (F_W + f) ds \\ F_w = \frac{1}{2}\rho A_D C_D v_j^2 \qquad v_j = \frac{v_i + v_{i+1}}{2} \\ f = KN \end{cases} \tag{7}$$

The track is divided into nine segments for dynamic analysis.

[segment1]

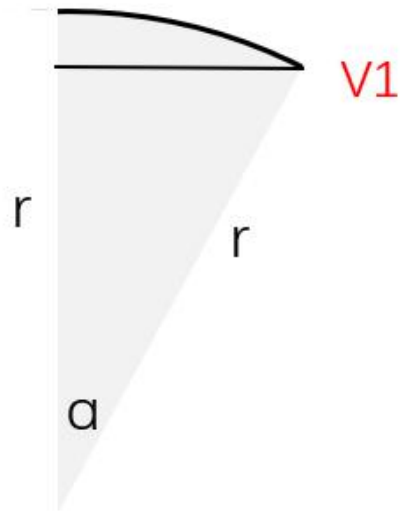


Figure 7 segment1

$$\begin{cases} mg \bullet \cos\alpha - W_1 = \frac{1}{2}mv^2 - 0 \\ W_1 = \int (F_w + f) dS = r\alpha \bullet F_w + r\alpha \int_0^{\frac{\pi}{2}} mg \bullet \cos\theta d\theta \end{cases} \tag{8}$$

[segment2]

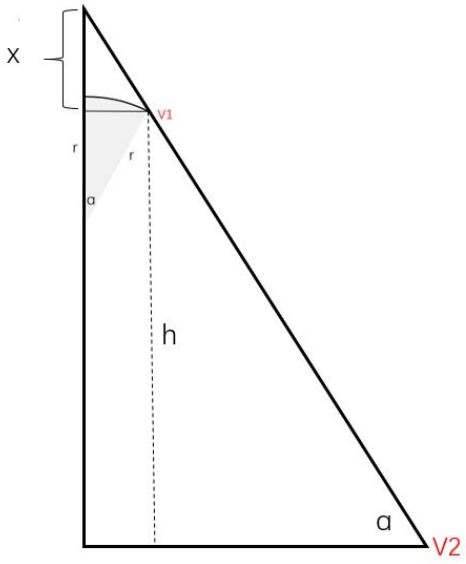


Figure 8 segment2

$$\begin{cases} \dfrac{h_2}{h_1+\left(27+1.5\sqrt{3}\right)}=\dfrac{8.5}{10} \\ \dfrac{h_1}{h_1+\left(27+1.5\sqrt{3}\right)}=\dfrac{1.5}{10} \\ mgh-W_2=\dfrac{1}{2}mv_2^2-\dfrac{1}{2}mv_1^2 \\ v_1=\dfrac{v_{1+}v_2}{2} \\ mgh-(mg\bullet\cos\alpha\bullet K\dfrac{8.5}{\cos\alpha}+F_w\dfrac{8.5}{\cos\alpha})=\dfrac{1}{2}mv_2^2-\dfrac{1}{2}mv_1^2 \end{cases} \tag{9}$$

[segment3]

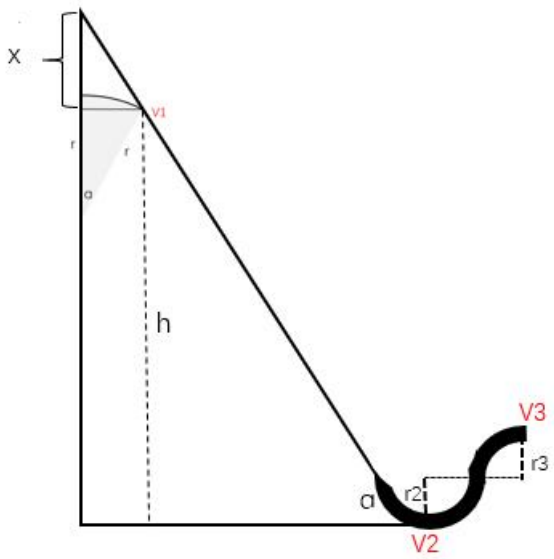


Figure 9 segment3

$$\begin{cases} -mg(\mathrm{r}_2+\mathrm{r}_3)-\mathrm{W}_3=\frac{1}{2}mv_3^2-\frac{1}{2}mv_2^2 \\ r_2=r_3 \\ W_3=0 \\ v_{\mathrm{II}}=\frac{v_{2+}v_3}{2} \end{cases} \tag{10}$$

[segment4]

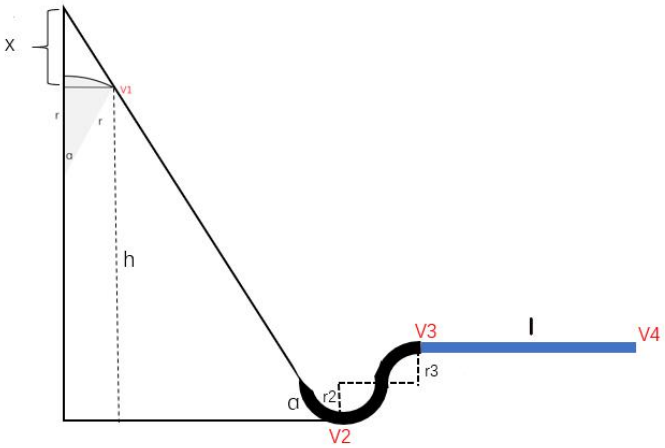


Figure 10 segment4

$$\begin{cases} -W_4=\frac{1}{2}mv_4^2-\frac{1}{2}mv_3^2 \\ W_4=-(\mathrm{kmg}+F_w)\bullet l \\ v_{\mathrm{III}}=\frac{v_{3+}v_4}{2} \end{cases} \tag{11}$$

[segment5]

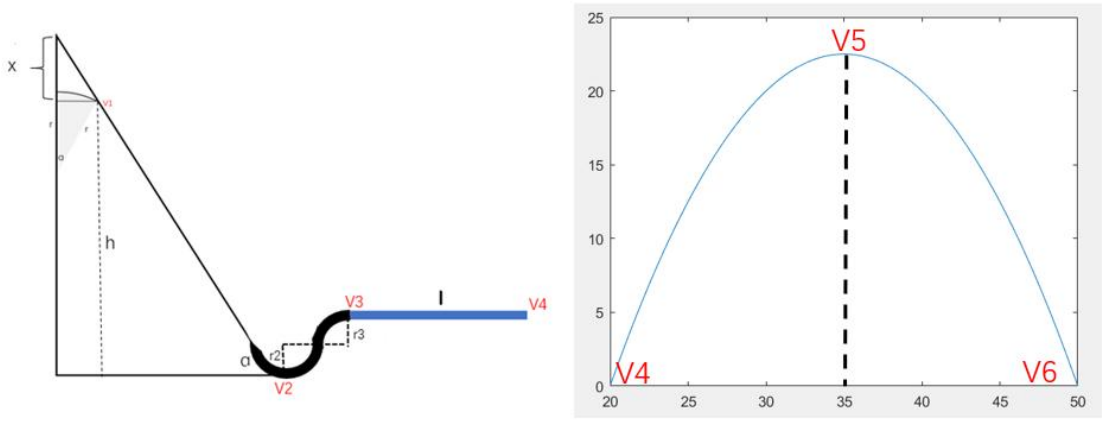


Figure 11 segment5

$$\begin{cases} y=-0.1\left[x-(10+\mathrm{r}_2+\mathrm{r}_3+l)\right]^2+22.5 \\ -W_5=\dfrac{1}{2}m(v_5^2-v_4^2) \\ W_{5_1}=\int\dfrac{1}{2}\rho \mathrm{A}_D\mathrm{C}_D{v_{\mathrm{IV}}}^2ds \\ W_{5_2}=\int\dfrac{1}{2}\rho \mathrm{A}_D\mathrm{C}_D{v_{\mathrm{V}}}^2ds \\ v_{\mathrm{IV}}=\dfrac{v_4+v_5}{2} \qquad v_{\mathrm{V}}=\dfrac{v_5+v_6}{2} \\ S=\int ds=\int_{20}^{50}\sqrt{1+0.04(\mathrm{x}-35)^2}dx \end{cases} \tag{12}$$

[segment6]

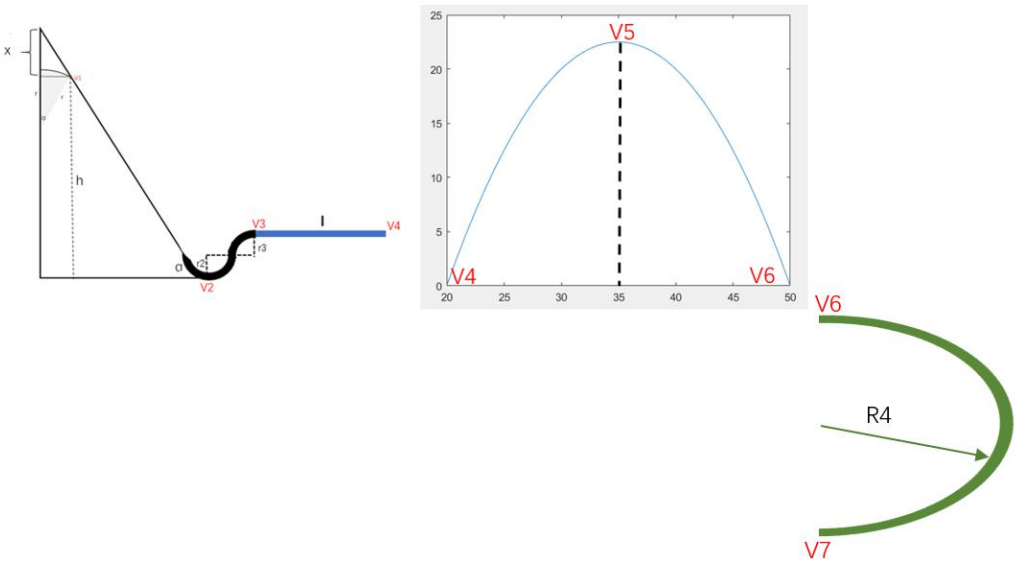


Figure 12 segment6

$$\begin{cases} -W_6 = \frac{1}{2}mv_7^2 - \frac{1}{2}mv_6^2 \\ W_6 = \pi \bullet r_4 \bullet (F_W + Kmg) \\ v_{VI} = \frac{v_6 + v_7}{2} \end{cases} \tag{13}$$

[segment7]

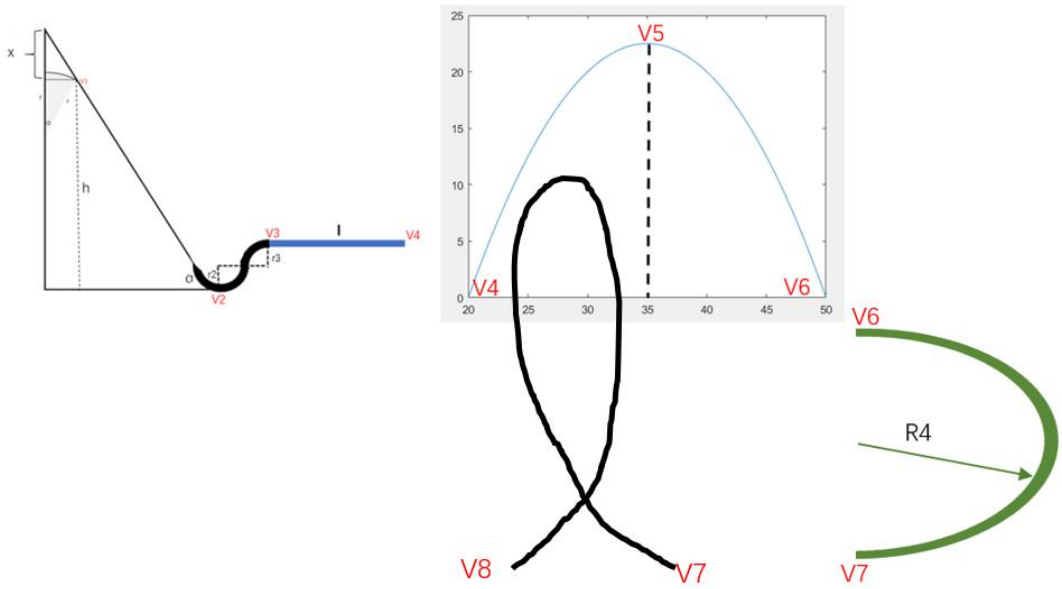


Figure 13 segment7

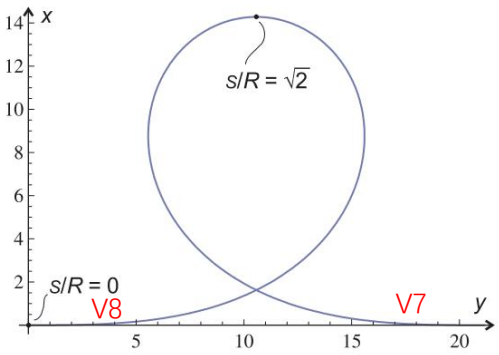
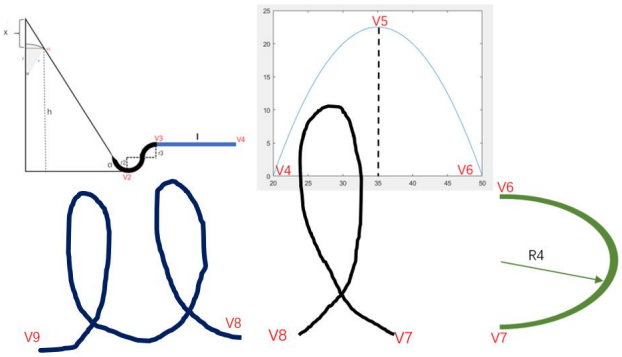


Figure 14 segment7_clothoid loop

$$v_7 = v_8 \tag{14}$$

[segment8]



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Figure 15 segment8

$$\begin{cases} x = a\cos\theta \\ y = a\sin\theta \\ z = b\theta \end{cases} \tag{15}$$

[segment9]

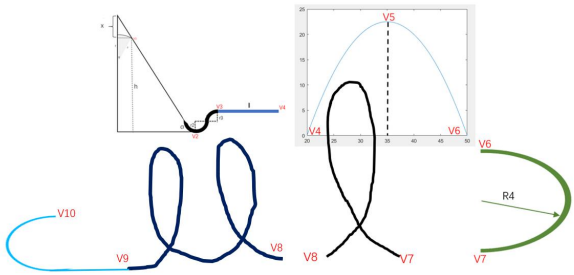


Figure 16 segment9

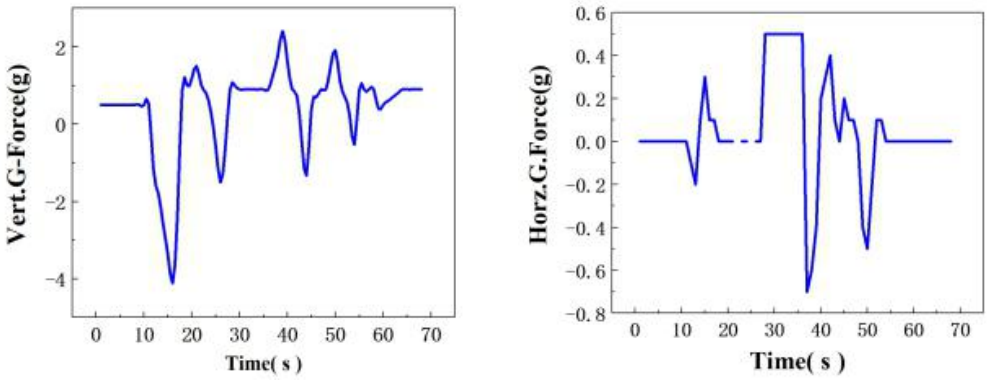
$$\begin{cases} -W_7 = \frac{1}{2}mv_{10}^2 - \frac{1}{2}mv_9^2 \\ W_7 = \left(\pi \bullet r_5 + l_2\right) \bullet \left(F_W + Kmg\right) \\ v_{10} \geq 0 \end{cases} \tag{16}$$

3. Results

3.1 Task 1

The thrill of a roller coaster is caused by being overweight on a cli mb and weightless on a descent, as well as by spinning and turning up side down.Under normal circumstances, a person would experience a vertical acceleration of 1g and would not experience a horizontal acce leration.While riding a roller coaster, however, being overweight or w eightless and changing direction can cause changes in acceleration, re sulting in a sense of excitement.However, excessive acceleration is bo und to cause damage to the health of passengers.Studies have shown t hat excessive and prolonged acceleration on roller coasters can signifi cantly increase the risk of injury.Normal adults can withstand 4~5g ve rtical acceleration with a duration of no more than 0.1s. Excessive acc eleration will make passengers' brain congestion.At the same time, ho rizontal acceleration exceeding 3g will greatly increase the possibility of spinal injury.Through the calculation of the theoretical model, we modify the design scheme, so as to obtain final design scheme, and

obtain the change of the acceleration of the scheme.



According to the theoretical calculation, the maximum vertical acceleration in this design scheme is 4.1g, and the duration is 0.01s, which occurs at the first downhill. The rest of the vertical acceleration is less than 4g, and the mean value is 0.4132g.The maximum theoretical value of transverse acceleration is 0.7g, which occurs in the rising part of the vertical ring. The time of overweight and weightlessness accounts for 87% of the total time. Therefore, this design scheme can continuously bring passengers a sense of tension and stimulation under the premise of ensuring safety.

3.2 Task 2

In the previous chapters, we have got some equations by anaylising the process of each part, and some data are defined and shown in tables. Now we just bring these data into equations and calculate them. Below are some critical results we have figured out.

Table 3 Instantaneous Velocity

Signal	Value(m/s)
V_1	4.486
V_2	24.31
V_3	19.874
V_4	20.28
V_5	20.16
V_6	20.04
V_7	15.32
V_8	15.32
V_9	15.32
V_{10}	10.37

From this table, we can see the velocity of the roller coaster in different positions. The table also roughly reflects the changing trend of its velocity.

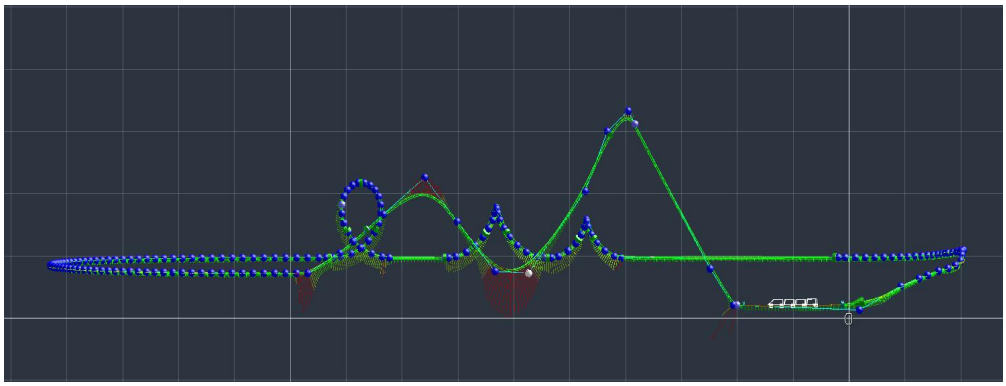


Figure 17 Instantaneous acceleration diagram

The full length of track and the total motion time of roller coaster are obtained by model simulation. The final result is shown in the table

below.

Table 4 total duration and length

Total duration	Lenth
01’02.67’’	518.16 meter

3.3 Task 3

According to *2.3 Model Establishment* as well as *2.4 Formula Derivation,* We can draw a simple diagram of the orbit,which connected with 9 segments. According to the schematic diagram of the track model and the derivation of the formula, the exact description of the roller coaster's motion along the track can be obtained.

Next, we will introduce the track of roller coaster in detail according to figure 4.

Step1: The roller coaster will begin with a 20 meter horizontal segment, where passengers board, then climb up a hill to its highest point, taking it to an altitude 30 meters above the starting location.

Step2: When the roller coaster is lifted to the top of the mountain, its instantaneous speed is zero, followed by a circle with a central Angle of 30° and a radius of 3 meters, followed by a straight line tangent to the circle, which is $22 + \sqrt{3}$ long.

Step3: It then slides into one half of the arc, then reaches the other end of the arc and moves upwards along a quarter of the arc with a central Angle of 45°. Then enter a straight section of 15 meters long.

Step4: The roller coaster goes up and then down along the parabola. After descending to the end of the parabola on the other side, it passes through a semi-arc with a radius of 50 meters. At this time, it moves in the opposite direction.

Step5: The roller coaster then follows a vertical loop at an altitude of 20 and a corkscrew formula.

Step6: Finally, after a section of the right curve back to the starting point, forming a closed loop.

4.Model Verification

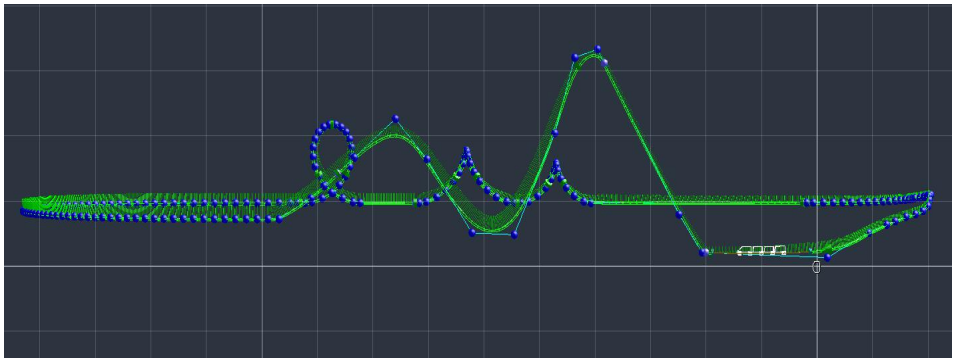


Figure 18 Instantaneous Velocity Image

Using Nolimts II: Roller Coaster Simulation software for the design of a Roller Coaster model kinematics Simulation, first of all in strict accordance with the design specifications of a Roller Coaster track modeling, using block design train of thought, each segment will orbit parameters such as dimension, Angle and height of the input, the orbital diagram.Next using simulator for the design of orbital kinematics simulation, according to the result of calculation and assumptions set

parameters such as the quality of the roller coaster, volume, and estimate its parameters, for the environment, with reference to experience wind resistance and friction coefficient, velocity and acceleration of the roller coaster is obtained by the simulator in runtime, etc., so as to verify the feasibility of the design, security, and stimulation, and comparing with the theoretical calculation value, institute set up a description of the mathematical model of roller coaster movement process and the stress distribution is feasible.

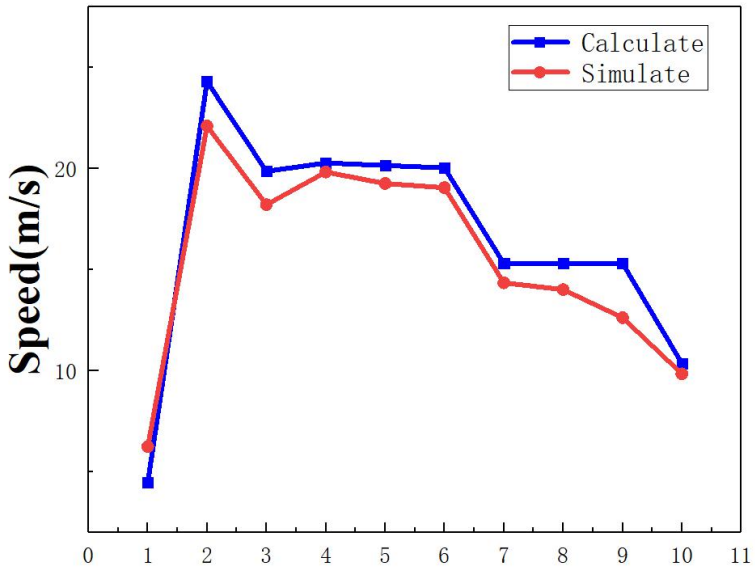


Figure 19 Speed comparison

The calculated average error is 2.1m/s, and the fitting degree of the two groups of data is as high as 94.26%. Therefore, it can be proved that the mathematical model proposed in this paper to describe the kinematic process of roller coaster is accurate and feasible in engineering application.

5.Model Improvement

The rolling friction could be calculated more precisely.

To make the model more closer to real situations, rolling friction in some sections of should be taken into consideration.

The average velocity in this paper is approximated as the arithmetic mean. However, the velocity of a roller coaster doesn't changes linearly, so we'll use other ways to calculate the average velocity.

We can do a more detailed analysis of the acceleration, so we could interpret the safety and excitement of the roller coaster more vividly and specifically.

6.Model Evaluation

Strength

The model in this paper shows analysis of each section of track in detail, and try to make out its variation during the whole process. It has not only the calculation in theory, but also a simulation of the roller coaster's motion. Through simulation, we can visually see the motion of roller coaster and evaluate its level of safety and excitement more easily and clearly.

Weakness

The analysis of the roller coaster's motion is still not thorough or precise enough. In some sections of its track, the situations are

simplified so greatly, that the results could have relatively much deviation from reality.

7.Conclusion

We analyse each section of the roller coaster's motion and figure out the velocity in theory in some positions. Then we know the rough variation of velocity during the motion. We simulate this motion so we see how it works in real situations. Comparing the velocity in theory and in real situations, we can see the model is relatively proper. By simulating the roller coaster's motion, we get the total duration of the ride. Through the changing of its acceleration, we conclude that the roller coaster is safe but especially exciting.

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9.Appendix

1.The theoretical and simulation values of the instantaneous velocity of the characteristic points

Signal	Value(m/s) Calculate	Value(m/s) Simulate
V_1	4.486	6.26
V_2	24.31	22.11
V_3	19.874	18.22
V_4	20.28	19.83
V_5	20.16	19.27
V_6	20.04	19.06
V_7	15.32	14.36
V_8	15.32	14.02
V_9	15.32	12.63
V_{10}	10.37	9.86

2.Data table of vertical and horizontal acceleration

1	0.5	0
2	0.5	0
3	0.5	0
4	0.5	0
5	0.5	0
6	0.5	0
7	0.5	0
8	0.5	0
9	0.5	0
10	0.5	0
11	0.5	0
12	-1.2	-0.1
13	-1.8	-0.2

14	-2.6	0.1
15	-3.4	0.3
16	-4.1	0.1
17	-2.3	0.1
18	0.9	0
19	1	0
20	1.2	0
21	1.5	0
22	1	0
23	0.8	0
24	0.3	0
25	-0.6	0
26	-1.5	0
27	-0.6	0
28	0.9	0.5
29	1	0.5
30	0.9	0.5
31	0.9	0.5
32	0.9	0.5
33	0.9	0.5
34	0.9	0.5
35	0.9	0.5
36	0.9	0.5
37	1.3	-0.7
38	1.8	-0.6
39	2.4	-0.4
40	1.6	0.2
41	0.9	0.3
42	0.7	0.4
43	-0.5	0.1
44	-1.3	0
45	0.4	0.2
46	0.7	0.1
47	0.9	0.1
48	0.9	0
49	1.5	-0.4
50	1.9	-0.5
51	1.1	-0.2
52	0.7	0.1
53	0.2	0.1
54	-0.5	0
55	0.9	0
56	0.9	0

57	0.9	0
58	0.9	0
59	0.4	0
60	0.5	0
61	0.6	0
62	0.7	0
63	0.8	0
64	0.9	0
65	0.9	0
66	0.9	0

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